Section:-A

1. Which development methodology would you recommend for this project, and why? Identify two specific risks the team would face if they chose the wrong methodology for this context.

Ans:

I would recommend the Agile methodology because the digital wallet project has frequently changing requirements. The client also wants to review working features every two weeks.

Agile develops software in small parts and allows the team to get feedback regularly. The team can make changes quickly based on the client's needs.

Two risks of choosing the wrong methodology:

1. **Late changes:** Using Waterfall may make changes difficult and costly after development has started.

2. **Client dissatisfaction:** The client may not get working features or feedback opportunities frequently enough.

2. Describe the Git workflow your team should adopt to prevent this from happening again. Name at least two Git commands your team should use as part of this workflow and explain the role of each command.

Ans:

The team should use a branch-based Git workflow. Every member should work on a separate branch instead of directly changing the main branch.

Steps:

1. Get the latest code using git pull.
2. Create a new branch using git checkout -b feature-name.
3. Make changes and save them using git add and git commit.
4. Push the branch using git push.
5. Create a pull request and ask a teammate to review the code.
6. Merge the code into the main branch only after review and testing.

Important Git commands:

- git pull — downloads and integrates the latest changes.
- git checkout -b feature-name — creates and switches to a new branch.
- git add . — selects changed files for the next commit.
- git commit -m "message" — saves a version of the changes.
- git push — uploads the branch to the remote repository.

3. List the appropriate HTML input types you would use for each of the five fields. Explain how you would use HTML5 built-in validation attributes to enforce the two

constraints mentioned, and why these attributes alone are not sufficient for production-level validation.

Ans:

The form should collect five fields:

Field	HTML input type
Applicant name	text
Email address	email
Phone number	tel
Preferred domain	select dropdown(Select one from Multiple option)
Resume upload	file

Explanation:

- required prevents submission when a field is empty.
- type="email" checks the basic email format.
- The dropdown limits the preferred domain to the listed options.
- JavaScript is not needed for these basic browser checks.

Why is this not enough for production?

Browser validation can be bypassed. The server must also validate the submitted data, check the resume file type and size, and verify that the information is safe and correct.

4. Explain how Bootstrap's grid class system works to produce this responsive layout. Write the Bootstrap class combination you would apply to the article column div, and explain what each class in the combination controls.

Ans:

Bootstrap uses a 12-column grid system. We can use different classes to control how many columns an article occupies at different screen sizes.

Use this class combination with Example:

```
<div class="col-12 col-md-4">
  Article content
</div>
```

Meaning:

- col-12 — uses all 12 columns on small screens, so articles appear one per row.
- col-md-4 — uses 4 out of 12 columns on medium and larger screens, so three articles fit in one row.

For a three-column desktop layout, place the article columns inside a Bootstrap .row container. Include the Bootstrap CSS file in the HTML page.

5. Explain why an array is the correct data structure for this task compared to using 30 separate variables. Describe the two-step logic your program would follow: first to compute the average, then to identify the above-average days — and explain why both steps cannot be done in a single loop.

Ans:

An array is useful because it stores all 30 rainfall readings under one variable name. We can access each day's reading using an index, such as rainfall[0] for the first day.

Using 30 separate variables would make calculations and data management more difficult.

Two-step logic:

1. **Calculate the average:** Add all 30 readings and divide the total by 30.
2. **Find above-average days:** Loop through the array again and print the day number whenever its rainfall is greater than the average.

Both steps are needed because the average must be calculated before we can compare each reading with it. A single ordinary forward pass cannot make these comparisons using the final average before that average is known.

6. Explain why the program crashes on an empty string input and describe the check your pointer-based solution must perform before dereferencing the pointer. How does traversing a string using a pointer differ from traversing it using array index notation, and which approach makes the empty-input bug easier to catch?

Ans:

An empty string contains the special character '\0' at its beginning. If a program tries to read beyond that terminator or uses an invalid pointer, it may crash or behave incorrectly.

Before processing a string, check whether the pointer is NULL. Then check whether the first character is '\0'.

Difference:

- **Array indexing:** str[i]
- **Pointer traversal:** *(ptr + i) or *ptr++

Both can traverse a string. Array indexing can make it easier for beginners to see the index and check whether the current character is '\0'. Pointer traversal also works safely when the pointer is checked and stops at the string terminator.